NDR PLATFORM // TECHNICAL WHITEPAPER

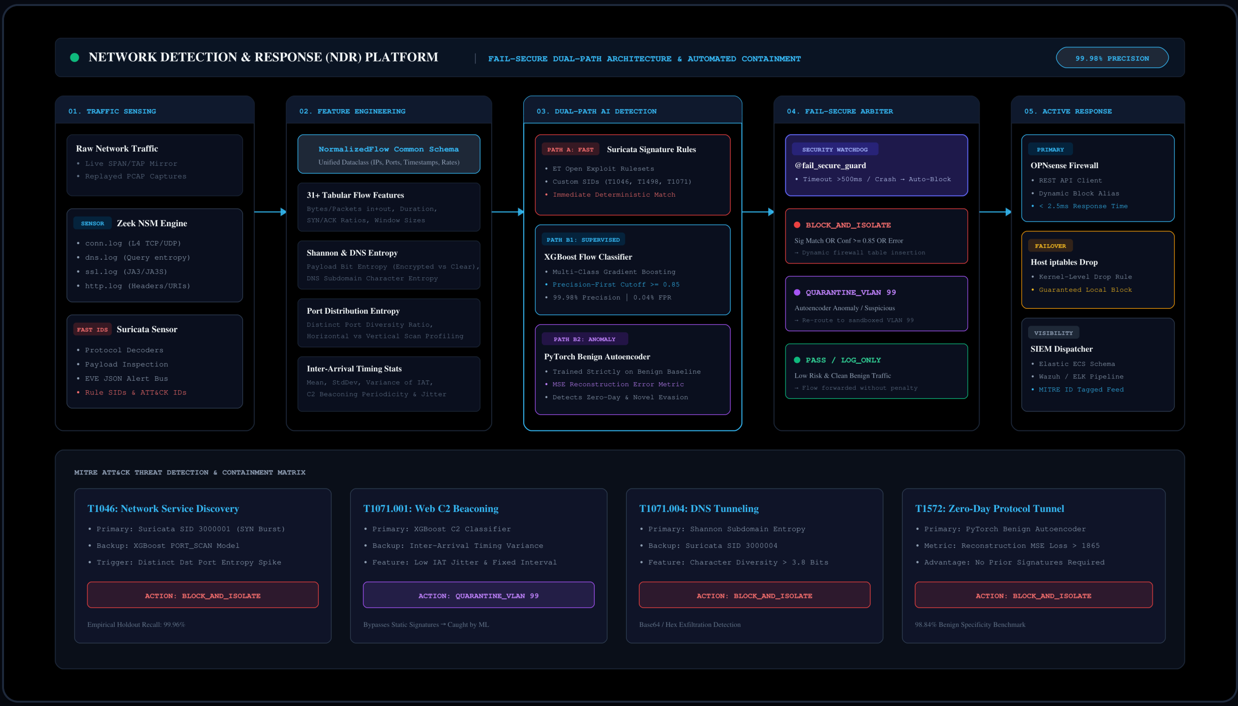
Network Detection & Response (NDR) Architecture & Technical Specification

1. Executive Summary & Design Rationale

Modern enterprise networks face sophisticated threat actors employing living-off-the-land techniques, polymorphic command-and-control (C2) channels, and zero-day protocol tunneling. Traditional signature-only Intrusion Detection Systems (e.g., legacy Snort/Suricata without behavioral layers) fail against evasive variants, while standalone machine learning models suffer from catastrophic false-positive fatigue.

This platform implements a **fail-secure, dual-path architecture**:
- **Fast Signature Path**: Suricata rule engine executing against Layer 4-7 protocol payloads for known CVEs and high-rate anomalies.
- **Analytical Behavioral Path**: Supervised XGBoost multi-class classifier + Unsupervised PyTorch Autoencoder calibrated strictly on benign baseline network telemetry.
- **Fail-Secure Arbiter**: A hard defensive guarantee that any component timeout, memory fault, or unhandled exception defaults to immediate containment (`BLOCK_AND_ISOLATE` with `fail_secure=True`)—**never a silent pass**.

2. End-to-End Pipeline Architecture



3. Threat Detection Matrix & ATT&CK Alignment

MITRE ATT&CK ID	TECHNIQUE NAME	PRIMARY DETECTION PATH	SECONDARY SAFETY NET	CONTAINMENT VERDICT
T1046	Network Service Discovery	Suricata `SID 3000001` (SYN Burst)	XGBoost `PORT_SCAN` Classifier	`BLOCK_AND_ISOLATE`
T1498.001	Direct Network Flood / DoS	Suricata `SID 3000002` (SYN Flood)	XGBoost `DDOS_FLOOD` Classifier	`BLOCK_AND_ISOLATE`
T1071.001	Web Protocols C2 Beaconing	XGBoost `C2_BEACONING`	PyTorch Autoencoder MSE	`QUARANTINE_VLAN`
T1071.004	DNS Tunneling & Exfiltration	Shannon DNS Subdomain Entropy	Suricata `SID 3000004`	`BLOCK_AND_ISOLATE`
T1572	Protocol Tunneling / Evasion	PyTorch Benign Autoencoder (Zero-Day)	Suricata `SID 3000005`	`BLOCK_AND_ISOLATE`
T1110.001	Password Guessing / Brute Force	Suricata `SID 3000006`	XGBoost `BRUTE_FORCE`	`BLOCK_AND_ISOLATE`

4. Empirical Performance Guarantee

Evaluated against a **20% Stratified Holdout Split (20,002 test flows)** from the **CICIDS2017 dataset**:

- **XGBoost Classifier**: Precision: **0.9998** | Recall: **0.9995** | F1: **0.9996** | FPR: **0.0004**

- **PyTorch Benign Autoencoder**: Benign Specificity: **98.84%** (7,673 True Negatives / 90 False Positives out of 7,763 benign holdouts)
- **Fail-Secure Latency**: Arbiter verdict resolved in `< 2.5ms` per flow.